reward signal, $r_t = r_i + r_e$. We can scale the intrinsic or extrinsic rewards to control the proportions of the total reward.

Figure 8.15. The DQN and the ICM contribute to a single overall loss function that is given to the optimizer to minimize with respect to the DQN and ICM parameters. The DQN's Q-value predictions are compared to the observed rewards. The observed rewards, however, are summed together with the ICM's prediction error to get a new reward value.

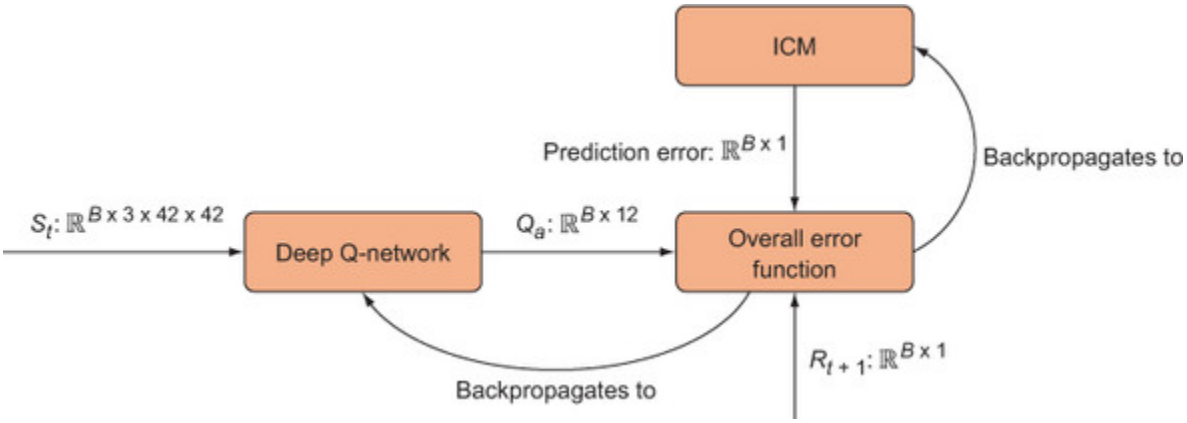


Figure 8.16 shows the ICM in more detail, including the agent model (DQN). Let's look at the code for the components of the ICM.

Figure 8.16. A complete view of the overall algorithm, including the ICM. First we generate B samples from the experience replay memory and use these for the ICM and DQN. We run the ICM forward to generate a prediction error, which is then provided to the DQN's error function. The DQN learns to predict action values that reflect not only extrinsic (environment provided) rewards but also an intrinsic (prediction error-based) reward.